Baloyi Thabang Bongani Junior

Quantitative Risk Analytics | Mathematical Statistics | Data Science & AI Engineering

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PROFESSIONAL SUMMARY

Mathematical statistician and quantitative-risk practitioner combining university teaching and research with industry experience in financial risk analytics, data science, and business analytics. MSc in Mathematical Statistics completed with distinction and a PhD in progress, supported by academic training in Statistics and Data Science, Actuarial Science, and quantitative finance.

Over three years of academic experience — lecturing undergraduate mathematics, running tutorials, setting and marking assessments, and supervising independent research — alongside more than three years of industry experience spanning actuarial pricing and reinsurance risk assessment, statistical and machine-learning modelling, model validation, optimisation, data engineering, and production AI systems for clients in banking, insurance, and defence.

Research interests align with business mathematics and informatics: deep-learning emulation and uncertainty quantification for insurance guarantee valuation, explainable machine learning for risk and pricing, hedging in incomplete markets, and Monte Carlo and optimisation methods. Brings established academic and professional networks, a growing publication record, and the ability to secure research funding and translate quantitative work into industry-directed projects.

EDUCATION

University of the Witwatersrand — Braamfontein, ZA
Doctor of Philosophy, Mathematical Statistics & Actuarial Sciences (Full Dissertation) — Jan 2027 – Present

- Research area: deep neural network emulation and uncertainty quantification for variable-annuity guarantee valuation under stochastic mortality

University of the Free State — Bloemfontein, ZA
Master of Science, Mathematical Statistics (Full Dissertation) — Jan 2026 – Oct 2026

- GPA: 80% (WITH DISTINCTION)
- Dissertation: "Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values"
- UFS Academic Record | MSc Dissertation

University of Cape Town — Cape Town, ZA
Bachelor of Science Honours, Statistics and Data Science, minor Actuarial Science — Graduated Dec 2025

- GPA: 74% | Major courses: Supervised & Unsupervised Learning, Multivariate Analysis, Statistical Computing, Actuarial Financial Economics, Portfolio Theory, Operational Research, Time Series, Bayesian Analysis
- Thesis: "Mean-Variance Hedging with Neural Networks using Vector AutoRegression (VAR) Models as the Underlying Sample Path Generator"
- UCT Transcript | Proof of Qualification

University of Cape Town — Cape Town, ZA
Bachelor of Commerce, Management Studies — Graduated Mar 2023

- GPA: 70% | BCom Certificate

University of the Free State — Bloemfontein, ZA
Bachelor of Science, Actuarial Sciences (Incomplete) — 2016 – 2017

- First-year coursework passed, several modules with distinction (record average 80%)

ACADEMIC & TEACHING EXPERIENCE

Lecturer (Mathematics) Aug 2025 – Dec 2025
Cape Peninsula University of Technology, Department of Maritime Studies Cape Town, ZA

- Took over the Mathematics (Vector Calculus I) module of the Bachelor of Marine Engineering programme mid-year, preparing a large cohort for the final examination
- Prepared lecture notes and tutorials; set the main and reassessment summative examinations; marked tests and examinations and submitted all marks on time
- Identified and closed learning gaps left after the previous lecturer's departure while completing the remaining syllabus under a constrained timeframe, delivering academic support often beyond standard hours

Teaching Assistant / Tutor 2022 – 2024
University of Cape Town, Department of Mathematics and Applied Mathematics Cape Town, ZA

- Ran weekly tutorial sections for first- and second-year mathematics and engineering mathematics courses, including calculus, linear algebra, vector calculus, and differential equations
- Invigilated tests and examinations, marked assessments, and provided academic support to undergraduate students across multiple courses
- Courses tutored include MAM1004F, MAM1012F, MAM1021S, MAM2012S, MAM2083F, MAM2084F/S, and MAM2085F

INDUSTRY EXPERIENCE

Senior Full Stack AI Engineer Jun 2026 – Present
4S4, Cape Town, ZA (Full Time, Hybrid)

- Clients: Rheinmetall Denel Munition, Old Mutual Bank, Ooba, Deloitte
- Led development of an enterprise demand and production-planning platform integrating SAP-derived demand, stock and supply, multi-level BOM explosion, finite work-centre capacity, lead-time calculation, inventory optimisation, and cross-plant scheduling
- Built explainable optimisation workflows using priority-based allocation, critical-path analysis, Monte Carlo lead-time simulation, linear programming, and lexicographic mixed-integer programming to identify capacity constraints and recommend feasible plans
- Develop and maintain data-integration pipelines (SAP S/4HANA-to-warehouse extraction, ETL, semantic layers) and orchestrate agentic, multi-agent, and RAG automation workflows

Artificial Intelligence Engineer Mar 2023 – May 2026
Full Stack, Cape Town, ZA (Full Time)

- Clients: Virgin Active (UK and SA), Expandly
- Designed and built AI-driven, cloud-native applications on AWS and Azure, integrating LLM services, data workflows, and Keras-based models into production systems
- Built data and ML workflows using Python, SQL, Spark, PostgreSQL, pgvector, Databricks, PySpark, and Delta Lake, supporting medallion-architecture data engineering and analytics-ready outputs
- Implemented CI/CD-supported testing, model validation, data-quality controls, and AI safety/compliance components, including guardrails and auditable decision workflows

Group Risk Intern (Actuarial) Jan 2021 – Dec 2022
Discovery Limited, Cape Town, ZA

- Supported actuarial pricing and risk assessment within the Group Risk reinsurance environment, analysing insurance and reinsurance exposures, claims, premiums, and risk data to support pricing adequacy
- Assessed reinsurance arrangements (retained and ceded risk, expected losses, exposure) and supported pricing calculations using actuarial and statistical methods, loss ratios, and expected-claims measures
- Performed sensitivity and scenario analysis, contributed to validation and review of actuarial model outputs and assumptions, and prepared analytical summaries for stakeholders

RESEARCH

MSc Dissertation — Jan 2026 – Oct 2026
University of the Free State, Department of Mathematical Statistics & Actuarial Science — Bloemfontein, ZA

- "Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values" — a bias-free deep neural network for health-insurance cost prediction, combining Monte Carlo simulation for input uncertainty quantification and Shapley values for model-agnostic interpretability
- Integrated a three-layer empirical framework (predictive, distributional, attributional) bridging deep-learning prediction with explainable actuarial modelling. Supervisor: Dr JM Blomerus
- GitHub Repository

BSc Honours Thesis — Feb – Nov 2024
University of Cape Town, Department of Statistical Sciences — Cape Town, ZA

- "Mean-Variance Hedging with Neural Networks using Vector AutoRegression (VAR) Models as the Underlying Sample Path Generator" — integrated neural networks with VAR models for mean-variance hedging in incomplete markets, demonstrating lower hedging errors and better risk-return trade-offs
- Supervised by Yuri Robbertze and Jake Stangroom. Repository

PUBLICATIONS

- Baloyi, T.B.J. "Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values." Manuscript in preparation, based on MSc dissertation, University of the Free State, 2026.

TECHNICAL & METHODOLOGICAL SKILLS

Programming: Python (TensorFlow, PyTorch, Keras, scikit-learn, NumPy, pandas, matplotlib), R, SQL, C#, MATLAB, STATA, Java, VBA, LaTeX
Risk, Statistical & Analytical: Financial Risk Analytics, Actuarial Pricing & Reinsurance, Statistical Modelling, Machine Learning, Deep Learning, Time Series, Multivariate & Bayesian Analysis, Stochastic Calculus, Portfolio Theory, Monte Carlo Simulation, Mixed-Integer & Linear Programming, Optimisation
Data, Cloud & Tools: PostgreSQL, Snowflake, Redshift, Delta Lake, dbt, pgvector, Spark/PySpark, Databricks, REST APIs, Flask, AWS, Azure, Docker, Git, GitHub Actions, Power BI, Tableau, Excel

PROFESSIONAL EXEMPTIONS & CERTIFICATIONS

Actuarial Society of South Africa (ASSA) exemptions: A112 — Business Economics; A113 — Business Finance
CFA Level I (In Progress)

HONOURS & AWARDS

- Dean's List, University of the Free State
- Best Aggregate Average Actuarial Science, University of the Free State & Moshal Scholarship
- Golden Key International Honour Society
- Moshal Scholarship | Jürgen Klauss Bathe Leadership Scholarship | UCT Postgraduate Funding Scholarship

ACADEMIC SERVICE & VOLUNTEERING

- UCT Mathematics Competition – Organiser / invigilator, 2019, 2023
- Tutor, Grade 8 and 9 – Adopt-a-school Foundation, 2021

REFERENCES

Ms Samkelisiwe Ntuli — ntulisa@cput.ac.za | +27 21 440 5769
Acting Head of Department, Department of Maritime Studies, Cape Peninsula University of Technology

Thomas van Heerden — thomas.vanheerden@uct.ac.za | +27 21 650 3200
Senior Lecturer, Department of Mathematics and Applied Mathematics, University of Cape Town

Reference letters available upon request.